Reviewer 1

Novelty: 9
Rationale: The idea of using a linguistic similarity matrix to form conceptual bridges when constructing prompts to improve cross-lingual transfer is one that I have not heard of before. I think this could be an interesting way of leveraging existing information about related languages for NLP tasks in general.

Feasibility: 8 (Highly Feasible: Straightforward to implement the idea and run all the experiments.)
Rationale: I think the idea makes sense, but more details should be shared about how exactly this language similarity matrix is constructed and what algorithms will be used for determining language similarity. More details should be provided on how the prompts for different languages will be obtained and how the data will be collected, which might be a time bottleneck.

Expected Effectiveness: 6 (Somewhat effective: There is a decent chance that the proposed idea can beat existing baselines by moderate margins on a few benchmarks.)
Rationale: I think that this idea could work well just by providing more context in different languages. The effectiveness sounds like it might be highly variable on the selection of pivot languages, though.

Excitement: 7
Rationale: I think that this could be interesting beyond the context of prompting, such as the use of pivot languages in traditional machine translation.

Overall Score: 7 (Good idea, would be accepted by major AI conferences)
Rationale: I think that the idea is sufficiently novel, and if it is executed well with good results, could produce a quality paper at a top NLP conference.

Confidence: 3 (You are fairly confident that the evaluation is correct)

Reviewer 2

Novelty: 8 (clearly novel - major differences from all existing ideas)
Rationale: The LPC method introduces a novel way of leveraging related languages and dialects to improve cross-lingual transfer. While cross-lingual transfer and language similarity have been explored, the idea of dynamically creating a constellation of prompts using pivot languages for specific tasks is a fresh and innovative approach.

Feasibility: 5 (Moderately feasible: It can probably be executed within the given time frame but would require careful planning, efficient use of APIs or some advanced computational strategies to overcome the limited GPU resources, and would require some modifications to the original proposal to make it work.)
Rationale: Implementing LPC could be challenging due to the complexities involved in selecting optimal pivot languages and designing effective prompts for each. While the concept is sound, the practical execution—such as building the language similarity matrix and dynamically generating prompts—may require substantial effort and experimentation.

Expected Effectiveness: 6 (Somewhat effective: There is a decent chance that the proposed idea can beat existing baselines by moderate margins on a few benchmarks.)
Rationale: The LPC method has the potential to improve cross-lingual performance, especially in low-resource languages. By leveraging linguistic similarities, the model might better understand and translate languages with limited training data.

Excitement: 7
Rationale: The LPC method is exciting because it tackles a critical challenge in multilingual NLP—improving performance for low-resource languages. If successful, it could significantly enhance the accessibility and usability of AI models across diverse linguistic contexts, particularly in underrepresented languages.

Overall Score: 6 (Marginally above the acceptance threshold of major AI conferences)
Rationale: The idea is a promising candidate for exploration in the field of multilingual NLP. It introduces a novel approach that could potentially improve cross-lingual transfer, particularly for low-resource languages and dialects. However, the challenges in implementation and the uncertain effectiveness of the method warrant a cautious overall rating.

Confidence: 4 (You are confident but not absolutely certain that the evaluation is correct)

Reviewer 3

Novelty: 8 (clearly novel - major differences from all existing ideas)
Rationale: Leveraging language similarity is often quite well studied in machine translation, but there hasn't been one studying using similar language as demonstration in multilingual in-context learning. It would be interesting to see how the model behavior change with different pivots.

Feasibility: 8 (Highly Feasible: Straightforward to implement the idea and run all the experiments.)
Rationale: The implementation will mostly involve building the similarity matrix and formatting the prompts. The similarity matrix should be able to get from some existing works. The prompt formatting and experiments part should be pretty straightforward with enough API quota.

Expected Effectiveness: 6 (Somewhat effective: There is a decent chance that the proposed idea can beat existing baselines by moderate margins on a few benchmarks.)
Rationale: The idea is pretty interesting, but it's not exactly sure whether similar languages are informative enough for the model, since it still requires the model to understand the similarity between languages and reason over the relationship between target language and the given languages.

Excitement: 8 (Exciting: would deepen the community's understanding or make major progress in this research direction)
Rationale: It would be informative to the community to see whether such demonstration can lead to good performance for in-context learning. Even if this idea doesn't work, the analysis will be quite informative.

Overall Score: 7 (Good idea, would be accepted by major AI conferences)
Rationale: This work studies an important problem for the multilingual community. The experiment results and analysis will be quite informative for multilingual in-context learning.

Confidence: 4 (You are confident but not absolutely certain that the evaluation is correct)

T Example Idea: LLM Directed Retrieval Querying for Improving Factuality

LLM Directed Retrieval Querying for Improving Factuality (Part 1)

1. Problem Statement: Large language models can generate flexible, long-form language generations, but LLM-generated responses often contain hallucinated or factually inconsistent content. Particularly in high-risk settings, there is a need for methods to improve the factuality of LLMs.

2. Motivation: A common framework for improving the factuality of LLM generations is retrieval augmented generation (RAG). In a RAG framework, a retriever takes a query as input and retrieves external knowledge from a high-quality knowledge base from reliable sources. The retrieved content is incorporated into the prompt for generating the response. One issue with this approach is that the quality of the generation can be bottlenecked by the quality of the retrieved content. Retrieval can be challenging for tasks where the query objective is underspecified or additional reasoning (or multi-step reasoning) on the query is required to retrieve content that supports the query.

3. Proposed Method: Our method refines the query by using an LLM to decompose the problem into sub-questions and generate candidate answers to expand each sub-question. The key steps include:

1. Decomposing the original question into sub-questions using an LLM.
2. Generating candidate answers for each sub-question using the LLM.
3. Expanding each sub-question with generated candidate answers to create retrieval queries.
4. Retrieving passages for each expanded query.
5. Filtering retrieved passages based on retrieval model score.
6. Aggregating filtered passages across sub-questions.
7. Prompting the generative LLM with the aggregated passages as context to answer the original question.

4. Step-by-Step Experiment Plan:

1. **Choose RAG datasets** where the retrieval task has underspecified/unique objectives or requires multi-hop reasoning, such as BIRCO and HotpotQA.
2. **Select a retriever,** such as an E5 or BGE model, and a generative LLM, such as GPT or LLaMA-3.
3. **Establish Baseline:**
   (a) Use the example question as the query to the retriever to retrieve relevant content from the retrieval passage pool.
   (b) Construct a prompt that provides the retrieved context passages and the question.
   (c) Prompt the generative LLM to answer the question using the context.
4. **Implement Proposed Method:**
   (a) Prompt the generative LLM to decompose the question into sub-questions.
   (b) For each sub-question, prompt the generative LLM to generate candidate answers.
   (c) Use semantic similarity to cluster the generated candidate answers and sample for semantic diversity.
   (d) Construct retrieval queries by expanding each sub-question with sampled candidate answers.
   (e) Retrieve passages using each query and aggregate results for each sub-question.
   (f) Deduplicate retrieved passages and filter based on retrieval model score.
   (g) Prompt the generative LLM with filtered passages as context to answer the original question.

LLM Directed Retrieval Querying for Improving Factuality (Part 2)

5. Test Case Examples:

- **Test Case 1:**
  - **Original Question:** In which region is the village after which lager "Fucking Hell" is named?
  - **Baseline:**
    * **Retrieval Query:** In which region is the village after which lager "Fucking Hell" is named?
    * **Retrieved Passage:** Fucking Hell is a German pale lager, a Pilsner, with an alcohol content of 4.9%. It is named after Fucking, the previous name of the village of Fugging in Austria; hell is the German word for 'pale' and a typical description of this kind of beer. The beer's name was initially controversial. Both the local authorities in Fucking and the European Union's Trade Marks and Designs Registration Office initially objected to the name. It was eventually accepted and the lager is sold internationally.
    * **Prompt:** Given the retrieved passage(s) as context and the question, answer the question using the context.
    * **Answer:** The village after which the lager "Fucking Hell" is named is located in Austria.
  - **Proposed Method:**
    * **Sub-Questions:**
      1. What village is the lager "Fucking Hell" named after?
      2. In which country is this village located?
      3. In which specific region or state within that country is the village located?
    * **Example Retrieval Query:** What village is the lager "Fucking Hell" named after? The lager "Fucking Hell" is named after the village previously known as Fucking, which is now called Fugging, in Austria.
    * **Retrieved Passages:**
      1. Fucking Hell is a German pale lager, a Pilsner, with an alcohol content of 4.9%. It is named after Fucking, the previous name of the village of Fugging in Austria; hell is the German word for 'pale' and a typical description of this kind of beer. The beer's name was initially controversial. Both the local authorities in Fucking and the European Union's Trade Marks and Designs Registration Office initially objected to the name. It was eventually accepted and the lager is sold internationally.
      2. Fugging, spelled Fucking until 2021, is an Austrian village in the municipality of Tarsdorf, located in the Innviertel region of western Upper Austria. It is 33 km (21 mi) north of Salzburg and 4 km (2.5 mi) east of the Inn river, which forms part of the German border.
    * **Prompt:** Given the retrieved passage(s) as context and the question, answer the question using the context.
    * **Answer:** The village after which the lager "Fucking Hell" is named is located in the Innviertel region of western Upper Austria.

6. Fallback Plan: If the proposed method does not satisfy the success criteria, alternative approaches could be explored. These may include quantifying the difficulty of various examples and analyzing whether this correlates with method improvement. The method is likely to be more effective for questions about esoteric facts, where the model is less likely to have internal knowledge of the answer, or its generated answers are more likely to disagree. Additionally, the method may be more beneficial for questions requiring information from multiple passages. Further analysis could help debug why the proposed method did not work, informing alternative new methods or transforming the project into an analysis paper by offering interesting ablations and insights.